

Bank Account Demonstration

Title: Bank Account Details using Classes and Objects

Objective/Aim:

To design and implement a Java program using classes and objects to manage bank account operations including account creation, deposit, withdrawal, interest calculation, and displaying account details.

System Requirements :

Hardware

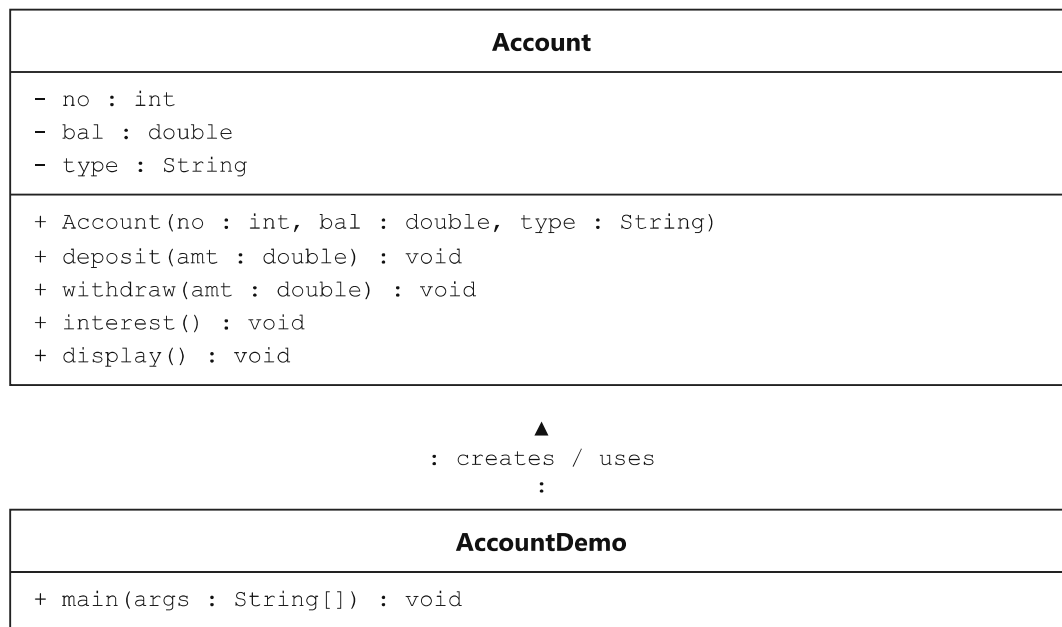
- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage
- Keyboard and Monitor

Software

- Operating System: Windows/Linux/macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt/Terminal

Algorithm / Logic Description:

1. Create a class named Account.
2. Define attributes such as:
 - no: Account number (integer)
 - bal: Account balance (double)
 - type: Account type (String)
3. Create a parameterized constructor to initialize the account details.
4. Define a `deposit(double amt)` method to add the amount to the balance and print confirmation.
5. Define a `withdraw(double amt)` method to check if balance is sufficient, deduct the amount if available, or display an insufficient balance error.
6. Define an `interest()` method to calculate and print 4% interest on current balance.
7. Define a `display()` method to display the account number, type, and current balance.
8. Create the AccountDemo class containing the `main()` method.
9. Create an Account object with:
 - Account Number: 1001
 - Balance: 10000.0
 - Type: Savings
10. Call the `display()`, `deposit()`, `withdraw()`, `interest()`, and `display()` methods.
11. Stop the program.

UML class Diagram

Relationship: AccountDemo creates objects and invokes methods of the Account class.

Source Code:

```

class Account {
    int no;
    double bal;
    String type;

    // Constructor to initialize account details
    Account(int no, double bal, String type) {
        this.no = no;
        this.bal = bal;
        this.type = type;
    }

    // Method to deposit money
    void deposit(double amt) {
        bal += amt;
        System.out.println("Deposited: " + amt);
    }

    // Method to withdraw money
    void withdraw(double amt) {
        if (amt <= bal) {
            bal -= amt;
            System.out.println("Withdrawn: " + amt);
        } else {
            System.out.println("Insufficient Balance");
        }
    }
}
  
```

```
// Method to calculate interest
void interest() {
    System.out.println("Interest: " + (bal * 0.04));
}

// Method to display account details
void display() {
    System.out.println("No: " + no);
    System.out.println("Type: " + type);
    System.out.println("Balance: " + bal);
}

}

public class AccountDemo {
    public static void main(String[] args) {
        // Creating an Account object
        Account a = new Account(1001, 10000, "Savings");

        // Performing banking operations
        a.display();
        a.deposit(2000);
        a.withdraw(3000);
        a.interest();
        a.display();
    }
}
```

Compilation & Execution Commands:

Save file as: AccountDemo.java

Compilation: javac AccountDemo.java

Execution: java AccountDemo

Output Screenshot:

```
PS C:\Users\mmeen> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' ... 'AccountDemo'
No: 1001
Type: Savings
Balance: 10000.0
Deposited: 2000.0
Withdrawn: 3000.0
Interest: 360.0
No: 1001
Type: Savings
Balance: 9000.0
PS C:\Users\mmeen>
```

Result:

The Bank Account Management program was successfully implemented in Java using classes and objects. The program demonstrates the use of a constructor, object creation, data members, and methods to deposit funds, withdraw funds, calculate interest, and display bank account details.